

ELL-HALER, Netherlands

WMO: 063770

Lat: 51.1967N

Lon: 5.7625E

Elev: 98

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.2	24.0	13.8	11.1	22.3	18.8	14.2	26.1	28.6	47.5	25.6	48.8	5.8	50	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.0	(c) 85.9	(d) 69.2	(e) 81.8	(f) 67.4	(g) 78.5	(h) 65.5	(i) 70.8	(j) 82.1	(k) 68.9	(l) 78.9	(m) 67.2	(n) 76.0	(o) 6.8	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.1	99.9	75.2	65.4	94.2	72.8	63.9	89.2	71.3	34.7	81.8	33.1	79.3	31.8	75.9	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.2	(b) 19.3	(c) 17.0	(e) 15.7	(f) 93.7	(g) 7.3	(h) 3.8	(i) 10.5	(j) 96.4	(k) 6.2	(l) 98.6	(m) 2.1	(n) 100.7	(o) -3.2	(p) 103.4		
(4) 22.2	(5) 19.3	(6) 17.0	(7) 15.0	(8) 75.5	(9) 7.1	(10) 1.8	(11) 9.9	(12) 76.9	(13) 5.7	(14) 77.9	(15) 1.8	(16) 79.0	(17) -3.4	(18) 80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	37.9	39.1	43.9	50.3	56.9	62.7	65.5	64.0	58.7	52.5	44.7	39.5	(6)
(7)		DBStd	11.66	7.80	7.26	6.44	6.56	6.32	5.58	5.60	4.82	5.11	6.38	6.22	7.41	(7)
(8)		HDD50	1543	379	309	206	77	12	0	0	0	1	48	181	330	(8)
(9)		HDD65	5163	841	726	655	442	262	108	64	76	201	389	609	790	(9)
(10)		CDD50	2041	3	3	16	86	225	382	479	434	261	126	21	5	(10)
(11)		CDD65	185	0	0	0	0	10	40	78	45	11	1	0	0	(11)
(12)		CDH74	1961	0	0	0	32	135	414	776	468	128	8	0	0	(12)
(13)		CDH80	541	0	0	0	2	23	105	241	147	23	0	0	0	(13)
(14)	Wind	WSAvg	7.9	9.9	9.5	8.8	7.7	7.6	6.9	6.7	6.4	6.3	7.5	8.2	9.4	(14)
(15)	Precipitation	PrecAvg	30.4	2.6	2.4	1.9	1.7	2.3	2.8	3.0	3.2	2.4	2.5	2.7	3.0	(15)
(16)		PrecMax	38.5	5.2	5.8	3.8	3.8	4.1	7.8	6.6	7.1	6.1	6.1	4.5	6.0	(16)
(17)		PrecMin	24.1	0.2	0.4	0.1	0.0	0.7	0.5	1.2	0.6	0.5	1.3	0.3	1.1	(17)
(18)		PrecStd	3.7	1.2	1.2	1.0	0.9	1.0	1.5	1.4	1.9	1.5	1.0	1.0	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	58.9	67.4	79.3	83.0	88.4	92.3	90.5	83.5	74.2	63.4	57.1	(19)
(20)		0.4%	MCWB	51.8	50.2	54.1	60.6	66.0	70.6	71.1	71.2	67.5	63.9	57.9	53.2	(20)
(21)		2%	DB	53.4	54.2	62.3	72.9	77.5	82.9	86.1	84.1	78.1	68.4	59.2	54.5	(21)
(22)		2%	MCWB	50.7	49.4	52.2	57.7	61.9	67.5	69.4	69.2	65.9	61.3	55.3	51.5	(22)
(23)		5%	DB	50.8	51.7	57.9	68.1	73.7	78.8	81.8	79.0	73.4	64.6	56.6	52.2	(23)
(24)		5%	MCWB	48.2	48.0	50.8	55.8	60.3	65.5	67.6	66.6	63.8	59.2	53.4	49.7	(24)
(25)		10%	DB	48.4	49.4	54.4	63.4	69.9	74.9	78.1	74.8	69.2	61.8	54.2	49.6	(25)
(26)		10%	MCWB	46.1	46.6	48.8	53.6	58.8	63.8	66.1	64.7	61.2	57.4	51.5	47.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.2	55.8	61.5	68.2	73.6	73.4	73.6	69.7	64.5	58.2	54.2	(27)
(28)		0.4%	MCDWB	54.6	55.5	63.7	74.5	79.9	85.4	87.2	85.9	79.5	72.1	61.8	56.3	(28)
(29)		2%	WB	50.7	50.3	53.5	58.9	64.6	69.3	70.8	70.2	66.9	61.7	55.7	51.9	(29)
(30)		2%	MCDWB	52.9	53.1	59.6	70.5	73.3	79.0	82.5	80.8	75.7	67.1	58.6	54.0	(30)
(31)		5%	WB	48.4	48.7	51.4	56.7	62.0	67.1	69.0	68.1	64.6	59.9	53.8	49.8	(31)
(32)		5%	MCDWB	50.6	51.6	56.7	66.0	70.2	75.7	79.1	77.4	71.8	64.1	56.3	52.0	(32)
(33)		10%	WB	46.2	46.4	49.4	54.5	59.8	65.1	67.2	66.1	62.5	58.0	51.5	47.7	(33)
(34)		10%	MCDWB	48.3	49.3	54.1	62.4	67.5	72.9	75.9	72.8	68.1	61.4	54.0	49.6	(34)
(35)	Mean Daily Temperature Range		MDBR	8.9	10.9	15.1	19.6	18.4	19.1	19.0	18.8	18.6	14.3	10.6	8.6	(35)
(36)		5% DB	MCDBR	9.6	14.3	22.5	28.1	25.6	26.1	26.8	26.9	25.0	17.7	12.4	9.8	(36)
(37)		5% DB	MCWBR	8.4	10.2	13.9	14.7	12.1	12.6	11.9	13.2	13.8	11.3	9.4	8.1	(37)
(38)		5% WB	MCDBR	9.8	12.8	19.6	24.8	21.6	22.5	23.6	23.5	22.8	16.9	11.5	9.3	(38)
(39)		5% WB	MCWBR	8.9	9.6	12.8	13.7	11.5	12.0	11.4	12.5	13.2	11.1	9.3	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.352	0.384	0.418	0.424	0.423	0.430	0.421	0.392	0.367	0.342	0.325	(40)	
(41)		taud	2.386	2.347	2.264	2.198	2.224	2.243	2.226	2.275	2.345	2.402	2.389	2.357	(41)	
(42)		Ebn at Noon	217	243	255	260	264	265	260	256	251	234	208	196	(42)	
(43)		Edh at Noon	20	27	35	41	42	42	42	38	32	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	239	454	819	1311	1571	1702	1606	1370	1016	598	281	184	(44)	
(45)		RadStd	32	88	119	146	140	145	194	129	104	50	32	25	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page